

For this question, your job will be to consider the likely results of a psychology experiment.

Imagine that 100 regular people are asked to judge which of two bags of sand is heavier, just by feeling the weight of the bags in their hands with their eyes closed.



The creators of the experiment have a “sand grain counter” and will use the numbers of grains of sand as a stand-in for the weight of each bag.



5000

Grains of sand

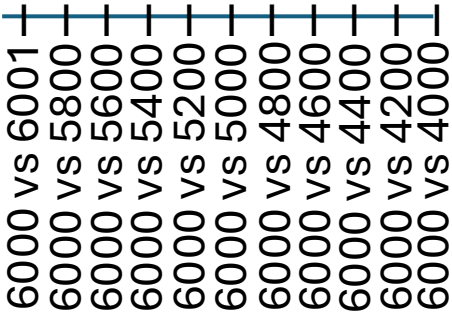


6000

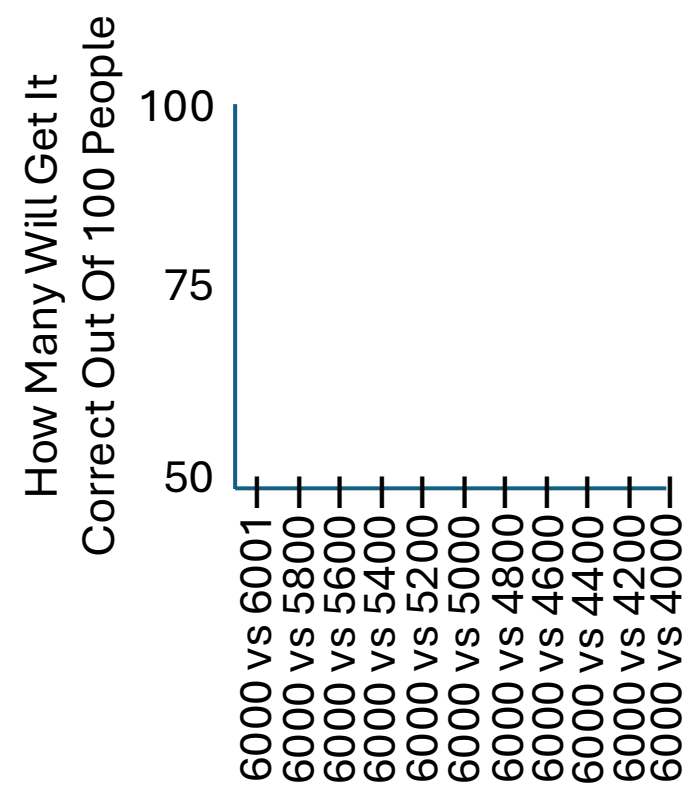
Grains of sand

You will be presented with 5 different graphs that show the possible outcomes from many such comparisons and your job will be to decide which one appears to be the correct graph.

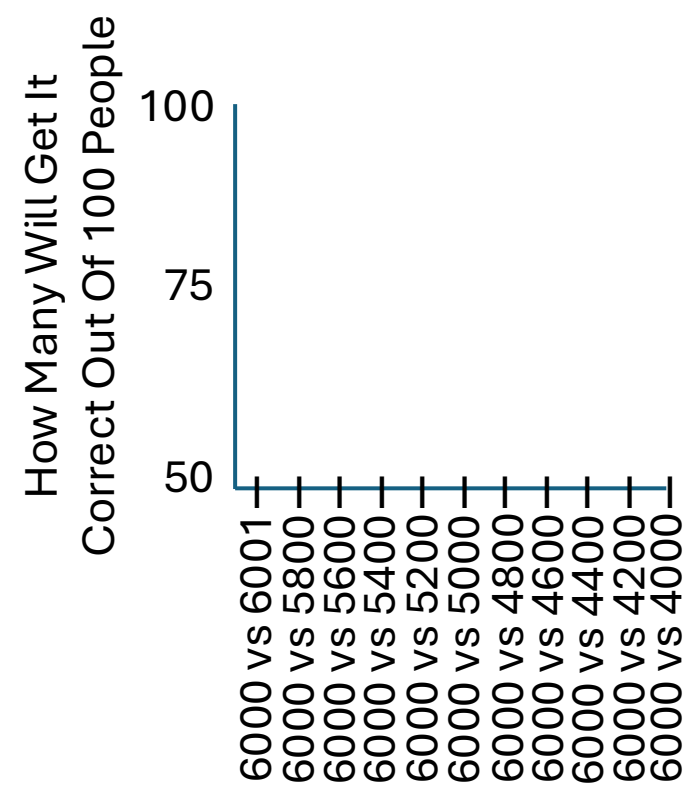
In each of the figures you will see, the bottom axis shows the numbers of grains to be compared. The easiest trials appear on the right (e.g., 6000 versus 4000 grains of sand) and the hardest trials on the left (e.g., 6000 versus 6001 grains of sand) with some relevant trials noted in between (e.g., 6000 versus 5600 grains of sand).



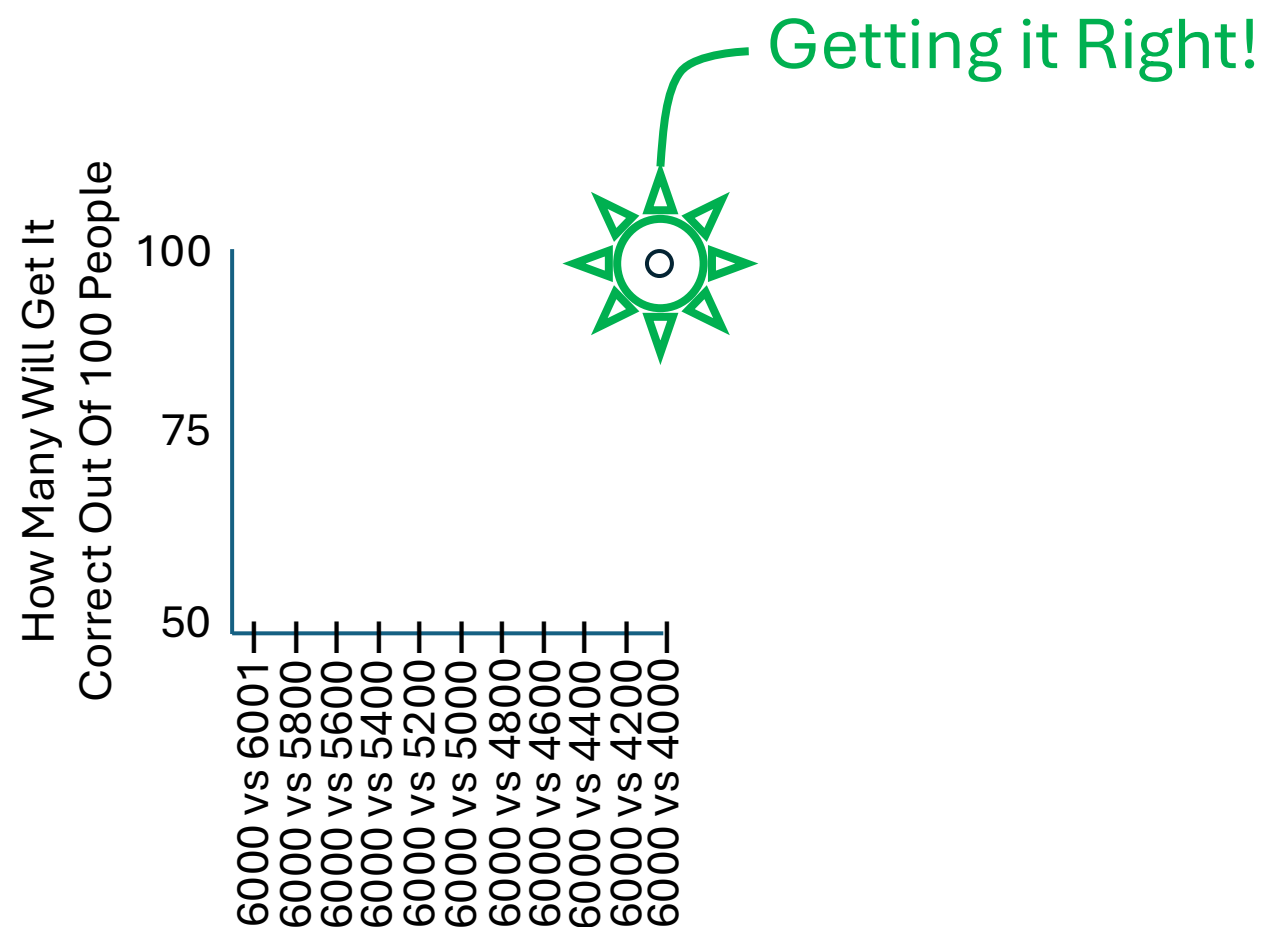
The vertical axis will show the number of people, out of 100, who will choose the correct bag of sand (i.e., with their eyes closes, will be able to feel which bag is heavier).



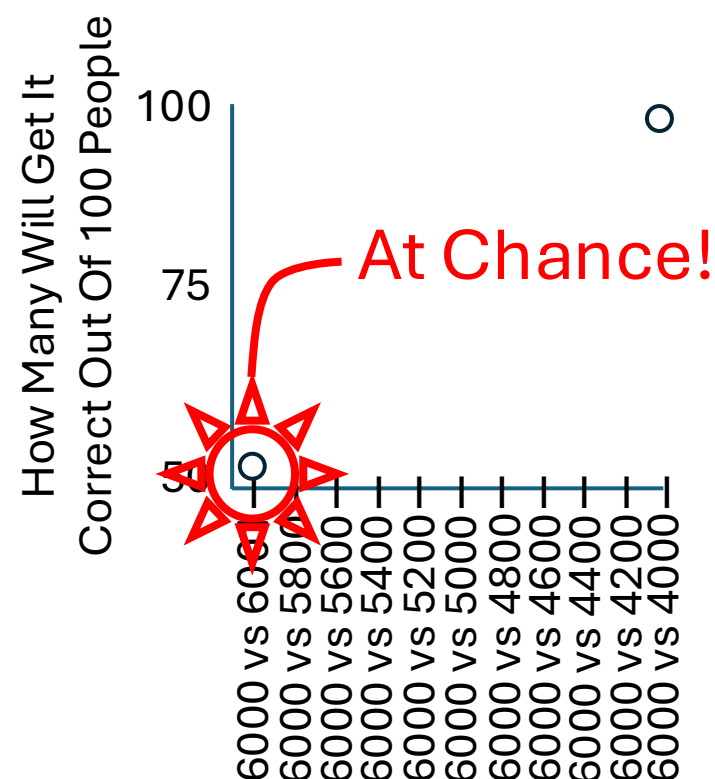
For each comparison, imagine attempting to do the trial yourself and decide “how many people out of 100 will choose the correct bag of sand”).



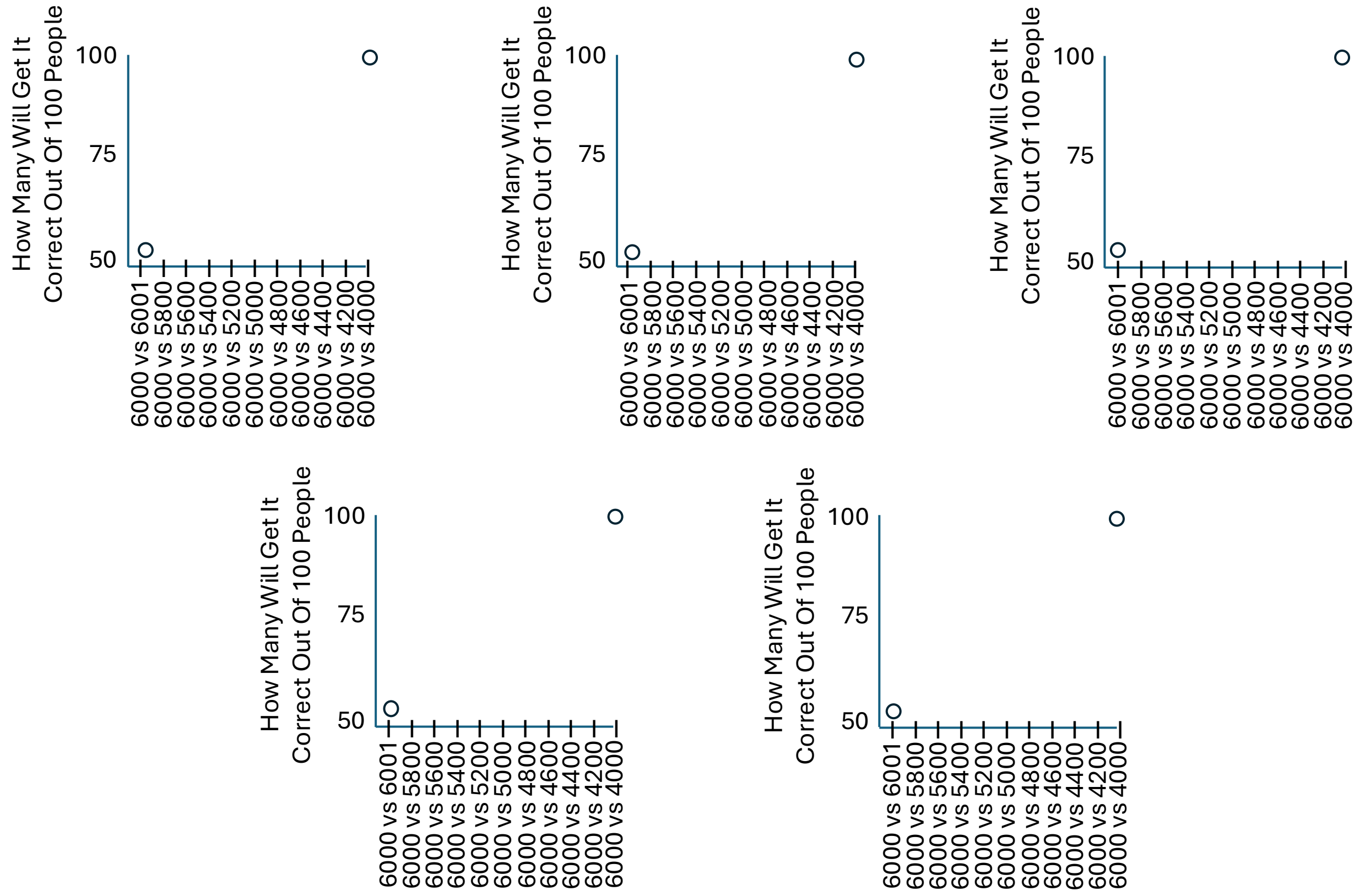
The creators of the experiment have provided two values to guide us. For the easiest trial (e.g., 6000 vs 4000), the creators estimate that “99 people out of 100 will get this correct” – because this trial is easy enough that nearly all people will successfully feel which bag is heavier.



For the hardest trial (e.g., 6000 vs 6001), the creators estimate that “51 people out of 100 will get this correct” – because this trial is so difficult that people are very close to chance performance. Notice that guessing would result in getting it correct half the time (just by chance) and getting it wrong half of the time – this is why the worst performance is at 50 people out of 100.



Now, imagine all of the possible combinations in between these two values. How do you expect the figure to look after many experimental trials of these sorts. We'll show you 5 possible graphs.



For each comparison dot, how many people out of 100 will choose the correct bag? Which pattern do you feel is the most likely to match the actual performance of regular human participants (choose one).

